



2010
Extension 2 MATHEMATICS
[March 2010]

YEAR 12
ASSESSMENT TASK 2

Time Allowed: 90 minutes + 5 minutes reading time

Topics: Complex Numbers

Name: _____

Teacher: _____

General Instructions:

- There are THREE (3) Questions which are of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.

$$\int \cos ax = \frac{1}{a} \sin ax + C$$

Total: 75 marks

Marks

QUESTION 1 (25 Marks)

- | | |
|--|---|
| a. Express $3cis\left(-\frac{\pi}{6}\right)$ in Cartesian form. | 2 |
| b. Solve $2x^2 + 3x + 4 = 0$ giving your answer in the form $x + iy$. | 2 |
| c. If $z = 3 + 2i$ and $w = 5 - i$, find in the form $x + iy$: | |
| i. $w - 2z$ | 2 |
| ii. wz | 2 |
| iii. $ w $ | 1 |
| iv. $\overline{z - w}$ | 2 |
| v. $z^2 + 2z + 1$ | 2 |
| d. Let $1, w, w^2$ be the cube roots of unity. | |
| i. Show that $1 + w + w^2 = 0$ | 2 |
| ii. Simplify $(1 + w)^3(3w + 3w^2)$ | 2 |
| e. Evaluate i^{3056} | 1 |

	Marks
f.	
i. Solve $z^5 = 1$ over the complex field.	2
ii. Plot your solutions on an Argand diagram	2
g. Simplify $\frac{\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)\left(\cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}\right)}{\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}}$ giving your answer in the form $a + bi$	3

QUESTION 2 (25 Marks)

a.	
i. Find the exact value of the modulus and argument of $\frac{1+2i}{1-3i}$	3
ii. Hence find the exact value of $\left(\frac{1+2i}{1-3i}\right)^9$ in Cartesian form.	2
b.	
i. Find $\sqrt{-3-4i}$	3
ii. Hence or otherwise, solve $z^2 - 3z + (3+i) = 0$	2

	Marks
c. The points A and B represent the complex numbers $4-i$ and $1+3i$ respectively.	
i. Plot the points A and B on an Argand diagram and mark the point P so that $OAPB$ is a parallelogram.	2
ii. What complex number does P represent?	1
d. If z is such that $ z = 3$ and $\arg z = \frac{2\pi}{3}$. Mark on the same Argand diagram:	
i. z	1
ii. $\bar{z}$	1
iii. $i\bar{z}$	1
iv. $2z$	1
v. $z - \bar{z}$	1
e. The roots of the equation $z^6 - 1 = 0$ are: $1, -1, \operatorname{cis}\frac{\pi}{3}, \operatorname{cis}\frac{2\pi}{3}, \operatorname{cis}\frac{-\pi}{3}, \operatorname{cis}\frac{-2\pi}{3}$. Use this information to express $z^6 - 1$ as a product of real quadratic factors	2
f. Factorise $z^3 - 8$ over the complex field.	2
g. Find an expression for $\cos 5\theta$ in terms of powers of $\cos \theta$	3

Marks

QUESTION 3 (25 Marks)

a. If $z = 1 + \sqrt{3}i$ and $|w| = 4$ find:

i. $|z|$

1

ii. the greatest value of $|z + w|$.

1

iii. if $|z + w|$ takes its maximum value, express w in the form $x + iy$

1

b. If $z = \cos \theta + i \sin \theta$

i. Show that $z^n - \frac{1}{z^n} = 2i \sin(n\theta)$

2

ii. Use the result obtained in i. to show that $\sin^4 \theta = \frac{1}{8}(\cos 4\theta - 4 \cos 2\theta + 3)$

2

iii. Hence evaluate $\int_0^{\frac{\pi}{2}} \sin^4 x dx$

2

c. Find the equation of the locus of each of the following in Cartesian form.

i. $|z - 1| = \operatorname{Re}(z)$

2

ii. $|z + 2 - i| = 4$

2

d. Sketch the locus of $|z - 1| = |z + i|$

2

Marks

e.

i. Sketch the locus given by $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{6}$

2

ii. Find the cartesian equation of the locus found in part i.

2

f. Sketch the following regions on separate Argand diagrams.

i. $|z + 2i| \leq 3$

3

ii. $|z - \bar{z}| < 2$ and $-\frac{\pi}{4} \leq \arg z \leq \frac{\pi}{4}$

3

End of Task 😊

Extension 2 Solutions:

Complex Numbers 2010

Question 1

$$\begin{aligned} \text{a. } 3 \cos\left(-\frac{\pi}{6}\right) + 3i \sin\left(-\frac{\pi}{6}\right) \\ = \frac{3\sqrt{3}}{2} - \frac{3}{2}i \end{aligned}$$

$$\begin{aligned} \text{b. } x &= \frac{-3 \pm \sqrt{3^2 - 4 \times 2 \times 4}}{2 \times 2} \\ &= \frac{-3 \pm \sqrt{-23}}{4} \\ &= \frac{-3 \pm \sqrt{23}i}{4} \end{aligned}$$

c.

$$\text{i. } 5 - i - 2(3 + 2i) = -1 - 5i$$

$$\begin{aligned} \text{ii. } (3 + 2i)(5 - i) &= 15 - 3i + 10i + 2 \\ &= 17 + 7i \end{aligned}$$

$$\text{iii. } \sqrt{5^2 + (-1)^2} = \sqrt{26}$$

$$\text{iv. } 3 - 2i - (5 + i) = -2 - 3i$$

$$\begin{aligned} \text{v. } (z + 1)^2 &= (3 + 2i + 1)^2 \\ &= (4 + 2i)^2 \\ &= 16 + 16i + 4i^2 \\ &= 12 + 16i \end{aligned}$$

d.

$$\begin{aligned} \text{i. } w^3 + w + w^2 &= w(w^2 + 1 + w) \\ w &\neq 0 \\ \therefore 1 + w + w^2 &= 0 \end{aligned}$$

$$\begin{aligned} \text{ii. } (-w^2)^3 (-3) &= -1 \times -3 \\ &= 3 \end{aligned}$$

e.

$$\begin{aligned} (i^4)^{764} &= 1^{764} \\ &= 1 \end{aligned}$$

f.

$$\text{i. } (rcis\theta)^5 = 1$$

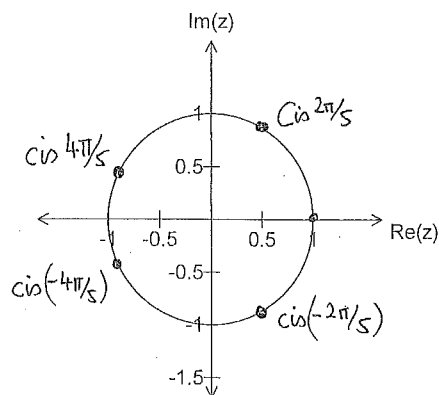
$$r^5 = 1, \quad r = 1$$

$$5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi$$

$$\theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, -\frac{4\pi}{5}, -\frac{2\pi}{5}$$

$$z = 1, cis\frac{2\pi}{5}, cis\frac{4\pi}{5}, cis\left(-\frac{4\pi}{5}\right), cis\left(-\frac{2\pi}{5}\right)$$

ii.



$$\begin{aligned} \text{g. } \frac{cis\frac{23\pi}{20}}{cis\frac{2\pi}{3}} &= cis\frac{29\pi}{60} \\ &= \cos\frac{29\pi}{60} + i \sin\frac{29\pi}{60} \\ &= 0.05233595624 + 0.9986295348i \end{aligned}$$

Question 2

$$\begin{aligned} \text{a. i. } \frac{1+2i}{1-3i} \times \frac{1+3i}{1+3i} &= \frac{1+5i+6i^2}{1-9i^2} \\ &= -\frac{1}{2} + \frac{1}{2}i \\ \text{mod} &= \frac{1}{\sqrt{2}} \text{ or } \frac{\sqrt{2}}{2} \\ \text{arg} &= \frac{3\pi}{4} \end{aligned}$$

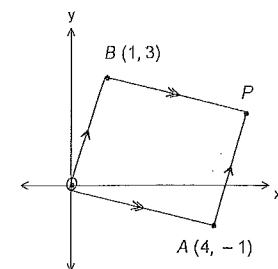
$$\begin{aligned} \text{a. ii. } \left(\frac{\sqrt{2}}{2} cis\frac{3\pi}{4}\right)^9 &= \frac{\sqrt{2}}{32} cis\left(\frac{27\pi}{4}\right) \\ &= -\frac{1}{32} + \frac{1}{32}i \end{aligned}$$

$$\begin{aligned} \text{b. i. } \sqrt{-3-4i} &= x + iy \\ x^2 - y^2 &= -3 \text{ and } 2xy = -4 \\ x^4 + 3x^2 - 4 &= 0 \end{aligned}$$

$$\begin{aligned} (x^2 + 4)(x^2 - 1) &= 0 \\ x &= \pm 1, \quad y = \mp 2 \\ \sqrt{-3-4i} &= \pm(1-2i) \end{aligned}$$

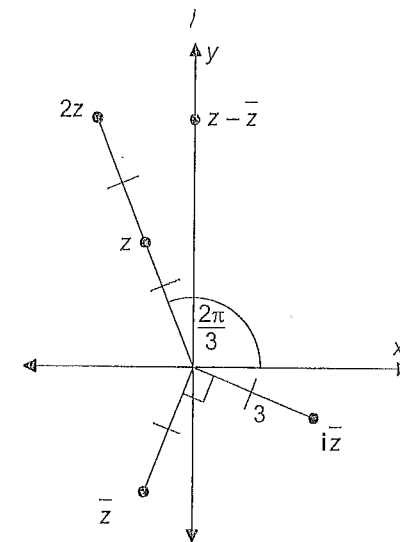
$$\begin{aligned} \text{b. ii. } z &= \frac{3 \pm \sqrt{9-4(3+i)}}{2} \\ &= \frac{3 \pm \sqrt{-3-4i}}{2} \\ &= \frac{3 \pm (1-2i)}{2} \\ z &= 2-i \text{ or } 1+i \end{aligned}$$

c. i.



$$\text{c. ii. } 5 + 2i$$

d.



$$\begin{aligned} \text{e. } (z-1)(z+1) &\left(z - cis\frac{\pi}{3}\right)\left(z - cis\frac{2\pi}{3}\right) \\ &\left(z - cis\left(-\frac{\pi}{3}\right)\right)\left(z - cis\left(-\frac{2\pi}{3}\right)\right) \\ &= (z^2 - 1)\left(z^2 - \left(2\cos\frac{\pi}{3}\right)z + 1\right)\left(z^2 - \left(2\cos\frac{\pi}{3}\right)z + 1\right) \\ &= (z^2 - 1)(z^2 - z + 1)(z^2 + z + 1) \end{aligned}$$

$$\begin{aligned} \text{f. } (z-2)(z^2 + 2z + 1 - 3i^2) \\ &= (z-2)((z+1)^2 - 3i^2) \\ &= (z-2)(z+1+\sqrt{3}i)(z+1-\sqrt{3}i) \end{aligned}$$

$$\begin{aligned} \text{g. } (\cos\theta + i\sin\theta)^5 &= c^5 + 5c^4is + 10c^3i^2s^2 + 10c^2i^3s^3 \\ &\quad + 5ci^4s^4 + i^5s^5 \\ \text{Equate real parts} \\ \cos 5\theta &= c^5 - 10c^3s^2 + 5cs^4 \\ &= c^5 - 10c^3(1-c^2) + 5c(1-c^2)^2 \\ &= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta \end{aligned}$$

Question 3

a.

$$i. |z| = \sqrt{1^2 + \sqrt{3}^2} = 2$$

$$ii. |z| + |w| = 2 + 4 = 6$$

iii. if $|z + w|$ takes its maximum value, then

$$\arg(z) = \arg(w).$$

$$\therefore w = 2z$$

$$w = 2(1 + \sqrt{3}i) = 2 + 2\sqrt{3}i$$

b.

$$i. z = \cos \theta + i \sin \theta$$

$$z^n = \cos(n\theta) + i \sin(n\theta)$$

$$z^{-n} = \cos(-n\theta) + i \sin(-n\theta) = \cos(n\theta) - i \sin(n\theta)$$

$$z^n - \frac{1}{z^n} = \cos(n\theta) + i \sin(n\theta) - (\cos(n\theta) - i \sin(n\theta)) = 2i \sin n\theta$$

$$ii. \text{ from i. } z - \frac{1}{z} = 2i \sin \theta$$

$$(2i \sin \theta)^4 = \left(z - \frac{1}{z}\right)^4$$

$$16 \sin^4 \theta = z^4 - 4z^3 \cdot \frac{1}{z} + 6z^2 \cdot \frac{1}{z^2} - 4z \cdot \frac{1}{z^3} + \frac{1}{z^4}$$

$$= \left(z^4 + \frac{1}{z^4}\right) - 4\left(z^2 + \frac{1}{z^2}\right) + 6 = 2 \cos 4\theta - 8 \cos 2\theta + 6$$

$$\sin^4 \theta = \frac{1}{8}(\cos 4\theta - 4 \cos 2\theta + 3)$$

$$iii. \int_0^{\frac{\pi}{2}} \sin^4 x \, dx = \frac{1}{8} \int_0^{\frac{\pi}{2}} (\cos 4x - 4 \cos 2x + 3) \, dx$$

$$= \frac{1}{8} \left[\frac{1}{4} \sin 4x - 2 \sin 2x + 3x \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{8} \left(\frac{3\pi}{2} - 0 \right)$$

$$= \frac{3\pi}{16}$$

c.

$$i. |z - 1| = \operatorname{Re}(z)$$

$$|x + iy - 1| = x$$

$$\sqrt{(x-1)^2 + y^2} = x$$

$$(x-1)^2 + y^2 = x^2$$

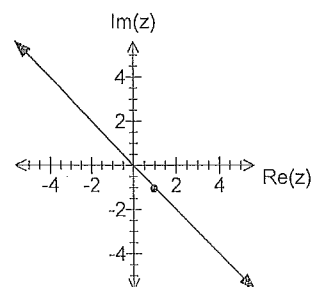
$$y^2 = 2x - 1$$

$$ii. |z + 2 - i| = 4$$

circle centre $(-2, 1)$ radius 4

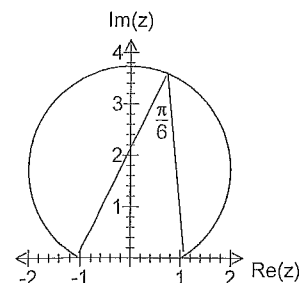
$$(x+2)^2 + (y-1)^2 = 16$$

d.

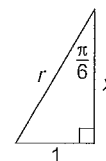


e.

i.



ii.



$$\tan \frac{\pi}{6} = \frac{1}{x}$$

$$x = \sqrt{3}$$

$\therefore$ centre of circle is $(0, \sqrt{3})$

By Pythagoras:

$$1 + \sqrt{3}^2 = r^2$$

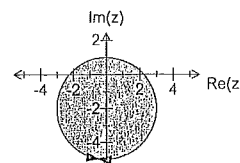
$$r = 2$$

$$\therefore \text{equation is: } x^2 + (y - \sqrt{3})^2 = 4$$

f.

i.

$$|z + 2i| \leq 3$$



ii.

$$|z + z| < 2 \text{ and } -\frac{\pi}{4} \leq \arg z \leq \frac{\pi}{4}$$

